# Acyclic component hypotheses for total cut complexes of disconnected graphs

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28 August 2026

*Internal successor manuscript v0.3-beta. The underlying proof and the complete Markdown manuscript have passed independent internal zero-trust audits. External mathematical review and journal peer review are pending.*

**Abstract**

Let $G = G_1 \sqcup \cdots \sqcup G_k$ be a finite simple graph with $k$ nonempty connected components and $n$ vertices. For $d \geq 2$, let $\Delta_d^t(G)$ be the total $d$-cut complex. We prove that if $k \geq d$ and, for every component $G_i$ and every $2 \leq r \leq d$, the complex $\Delta_r^t(G_i)$ is void or integer-acyclic, then

$$\Delta_d^t(G) \simeq \bigvee_{\binom{k-1}{d-1}} S^{n-d-1}.$$

This extends the author's earlier low-component preprint [8], which answered Question 29 of Carnero Bravo under the written hypotheses of the source theorem. The present result also answers Question 30 for the stated component class: it removes the simple-connectivity condition on the component clique complexes and weakens component contractibility to integer acyclicity throughout all $k \geq d$. The homology calculation replaces an objectwise-contractible homotopy-colimit argument by its integral homology spectral sequence. The remaining homotopy problem is closed by direct connectivity arguments. The most delicate layer, $k = d$, is handled by a weak-composition cover whose nonbaseline blocks are cones and whose relevant intersections are connected. The acyclicity assumption cannot simply be deleted: for $G = C_4 \sqcup K_1$ and $d = 2$, the total cut complex has an additional first homology class.

## 1 Introduction

For a graph $G$ and an integer $d \geq 1$, the total $d$-cut complex is

$$\Delta_d^t(G) = \{\sigma \subseteq V(G) : \alpha(G - \sigma) \geq d\}. \tag{1.1}$$

Equivalently, a vertex set is a face precisely when its complement contains an independent set of size $d$. Total cut complexes were introduced by Bayer, Denker, Jelić Milutinović, Rowlands, Sundaram, and Xue [1]. Their topology is closely related to the bounded-independence complexes

$$\mathrm{BI}_d(G) = \{\sigma \subseteq V(G) : \alpha(G[\sigma]) < d\}, \tag{1.2}$$

because $\Delta_d^t(G)$ and $\mathrm{BI}_d(G)$ are combinatorial Alexander duals.

Carnero Bravo [2] developed homotopy-colimit decompositions for these filtrations. In particular, [2, Theorem 28] proves the wedge formula

$$\Delta_d^t(G) \simeq \bigvee_{\binom{k-1}{d-1}} S^{n-d-1} \tag{1.3}$$

when $G$ has $k \geq d+3$ connected components, each component total-cut complex in the relevant range is void or contractible, and each component clique complex $\mathrm{BI}_2(G_i)$ is simply connected. The two questions immediately following that theorem ask whether (1.3) also holds for $k = d, d+1, d+2$ under the same hypotheses [2, Question 29], and whether the clique-complex connectivity condition can be removed [2, Question 30].

The author's earlier preprint [8] answered Question 29 under the written void-or-contractible and clique-complex hypotheses of [2, Theorem 28]. The present manuscript subsumes that result and resolves Question 30 on a slightly larger component class. Its new content relative to [8] is the acyclic composition-diagram argument, the corresponding removal of the component clique-complex condition, the $k = d$ connected-intersection mechanism under the weaker hypothesis, and the negative control in Section 7. If released, this manuscript is intended as a successor to [8], not as an unrelated claim of first priority for the three low-component layers.

*Theorem A.* Let $d \geq 2$, and let

$$G = G_1 \sqcup \cdots \sqcup G_k$$

be a finite simple graph with $k \geq d$ nonempty connected components and $n$ vertices. Suppose that, for every $i$ and every $2 \leq r \leq d$, the complex $\Delta_r^t(G_i)$ is either void or integer-acyclic. Then

$$\Delta_d^t(G) \simeq \bigvee_{\binom{k-1}{d-1}} S^{n-d-1}. \tag{1.4}$$

Here integer-acyclic means nonvoid with all reduced integral homology groups zero. Thus every contractible complex is integer-acyclic, but no contractibility or fundamental-group condition is assumed.

There are two separate issues in the proof. First, the composition-poset diagram of [2] still determines integral homology when its objects are merely integer-acyclic. The homology spectral sequence of the homotopy colimit collapses to the homology of the composition poset. Alexander duality then gives the homology of (1.4). Second, this homology must be upgraded to a homotopy equivalence. For $k \geq d+1$, the required simple connectivity is purely combinatorial and does not use any component clique-complex condition. For $k = d$, the complete-2-skeleton method is unavailable; a weak-composition cover and a van Kampen attachment argument replace it.

Theorem A formally weakens the written hypotheses of [2, Theorem 28], but we do not claim that integer acyclicity is strictly more general within every natural graph family. A finite search conducted during discovery found no small component total-cut complex that was integer-acyclic but noncontractible. That search is not part of the proof. On the other hand, Section 7 gives a small counterexample showing that the acyclicity hypothesis cannot be removed altogether.

## 2 Definitions and conventions

All graphs and simplicial complexes are finite. A complex $K$ is taken on a fixed finite ambient ground set $\Omega$ and may have ghost vertices, meaning elements of $\Omega$ whose singleton is not a face. We distinguish the void complex on $\Omega$, which has no faces, from $\{\varnothing\}$. We use $\{\varnothing\} = S^{-1}$ as the identity for joins. The combinatorial Alexander dual of $K$ relative to $\Omega$, where $N = |\Omega|$, is

$$K^* = \{\sigma \subseteq \Omega : \Omega - \sigma \notin K\}. \tag{2.1}$$

Combinatorial Alexander duality gives [3]

$$\widetilde{H}_j(K;\mathbb{Z}) \cong \widetilde{H}^{N-j-3}(K^*;\mathbb{Z}). \tag{2.2}$$

The complexes $\Delta_r^t(H)$ and $\mathrm{BI}_r(H)$ are both taken on the fixed ground set $V(H)$. Definitions (1.1) and (1.2) immediately imply

$$\Delta_r^t(H) = \mathrm{BI}_r(H)^*. \tag{2.3}$$

For weak-composition formulas we extend the total-cut filtration by

$$\Delta_0^t(H) = \Delta^{V(H)}, \qquad \Delta_1^t(H) = \partial\Delta^{V(H)}. \tag{2.4}$$

The second equality holds because $H - \sigma$ contains an independent vertex if and only if $\sigma \neq V(H)$. For bounded-independence joins we set

$$\mathrm{BI}_1(H) = \{\varnothing\}. \tag{2.5}$$

**Lemma 2.1.** *If deleting a vertex set $\sigma$ leaves at least $d$ nonempty connected components of $G$, then $\sigma \in \Delta_d^t(G)$.*

*Proof.* Choose one remaining vertex from each of $d$ components. Vertices in distinct components are pairwise nonadjacent. □

We also use the standard fact that a simply connected CW complex whose only nonzero reduced integral homology is $\mathbb{Z}^b$ in degree $q \geq 2$ is homotopy equivalent to a wedge of $b$ copies of $S^q$ [4].

## 3 The acyclic composition diagram

This section computes the integral homology of both dual complexes throughout the range $k \geq d$.

**Lemma 3.1.** *Under the hypotheses of Theorem A, for every $i$ and $2 \leq r \leq d$, the complex $\mathrm{BI}_r(G_i)$ is nonempty and integer-acyclic.*

*Proof.* If $\Delta_r^t(G_i)$ is void, its Alexander dual is the full simplex on $V(G_i)$. Suppose instead that $\Delta_r^t(G_i)$ is integer-acyclic. The integral universal coefficient theorem gives vanishing reduced integral cohomology, and (2.2) then gives vanishing reduced integral homology for $\mathrm{BI}_r(G_i)$. Since $r \geq 2$, the empty face and every singleton lie in $\mathrm{BI}_r(G_i)$, so it is nonempty. □

Let $m = d + k - 1$. Following [2], define the finite poset

$$\mathcal{C}_{m,k} = \bigcup_{s=k+1}^{m} \{(c_1, \dots, c_k) \in \mathbb{Z}^k_{\geq 1} : c_1 + \cdots + c_k = s\}, \tag{3.1}$$

ordered coordinatewise. For $\mathbf{c} \in \mathcal{C}_{m,k}$, put

$$X_{\mathbf{c}} = \mathrm{BI}_{c_1}(G_1) * \cdots * \mathrm{BI}_{c_k}(G_k). \tag{3.2}$$

Every coordinate satisfies $1 \leq c_i \leq d$. Moreover, at least one coordinate is at least 2.

**Lemma 3.2.** *Every $X_{\mathbf{c}}$ is nonempty, connected, and integer-acyclic.*

*Proof.* Choose a coordinate $i$ with $c_i \geq 2$. By Lemma 3.1, the corresponding factor is nonempty and integer-acyclic. Factors with $c_j = 1$ are join identities; all other genuine factors are nonempty. The reduced chain complex of a finite join satisfies the standard tensor–Tor join formula. A join with one integer-acyclic genuine factor therefore has zero reduced integral homology. It has vertices, so vanishing reduced $H_0$ also says that it is connected. □

The composition decomposition of [2, Lemma 25 and Corollary 27] gives

$$\mathrm{BI}_d(G) = \operatorname*{colim}_{\mathcal{C}_{m,k}} X_{\mathbf{c}} \simeq \operatorname*{hocolim}_{\mathcal{C}_{m,k}} X_{\mathbf{c}}. \tag{3.3}$$

The second equivalence is the standard homotopy-colimit comparison for this finite diagram of subcomplex inclusions; the diagram includes all its nonempty finite intersections.

**Proposition 3.3.** *Under the hypotheses of Theorem A,*

$$\widetilde{H}_j(\mathrm{BI}_d(G); \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{k-1}{d-1}}, & j = d-2, \\ 0, & j \neq d-2. \end{cases} \tag{3.4}$$

*Proof.* The integral homology spectral sequence for the homotopy colimit in (3.3) has

$$E^2_{p,q} = H_p(\mathcal{C}_{m,k}; \mathcal{H}_q(X)) \Longrightarrow H_{p+q}(\operatorname{hocolim} X; \mathbb{Z}). \tag{3.5}$$

By Lemma 3.2, $\mathcal{H}_q(X) = 0$ for $q > 0$. The $H_0$ coefficient system is the constant system $\mathbb{Z}$: every object is nonempty and connected, and every structure inclusion induces the identity between its canonical $H_0 \cong \mathbb{Z}$. Thus (3.5) has only its $q = 0$ row and yields

$$H_*(\mathrm{BI}_d(G); \mathbb{Z}) \cong H_*(\Delta(\mathcal{C}_{m,k}); \mathbb{Z}). \tag{3.6}$$

By [2, Proposition 12], for $k \geq d$ the order complex is homotopy equivalent to

$$\Delta(\mathcal{C}_{d+k-1,k}) \simeq \bigvee_{\binom{k-1}{d-1}} S^{d-2}. \tag{3.7}$$

Equations (3.6) and (3.7) prove the claim. □

**Corollary 3.4.** *Under the hypotheses of Theorem A,*

$$\widetilde{H}_j(\Delta^t_d(G); \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{k-1}{d-1}}, & j = n-d-1, \\ 0, & j \neq n-d-1. \end{cases} \tag{3.8}$$

*Proof.* Apply combinatorial Alexander duality to (3.4). Since the single nonzero group is free, the integral universal coefficient theorem introduces neither hidden torsion nor extension terms. If $n = k = d$, the target degree is $-1$ and the statement follows directly from $\Delta^t_d(dK_1) = \{\varnothing\} = S^{-1}$. □

## 4 Connectivity for $k \geq d+1$

Put

$$q = n - d - 1. \tag{4.1}$$

Every face of $\Delta_d^t(G)$ has at most $n-d$ vertices, because its complement contains an independent $d$-set. Hence

$$\dim \Delta_d^t(G) \leq q. \tag{4.2}$$

The low-dimensional cases can therefore be read directly from (3.8): if $q = 0$, the complex is a set of $\binom{k-1}{d-1}+1$ points; if $q = 1$, it is a connected graph with cycle rank $\binom{k-1}{d-1}$ and hence the required wedge of circles. We now establish simple connectivity whenever $q \geq 2$.

**Proposition 4.1** (The layer $k = d+1$). *If $k = d+1$, then*

$$\Delta_d^t(G) \simeq \bigvee_d S^{n-d-1}. \tag{4.3}$$

*Proof.* Let $e = n - k = q$. The cases $e = 0, 1$ are the low-dimensional cases just described. For completeness, when $e = 0$ the complex consists of $d+1$ points. When $e = 1$, one component is a copy of $K_2$ and all others are singletons; the complex is obtained from $K_{2,d}$ by adding the edge inside the $K_2$, and its cycle rank is $d$.

Assume $e \geq 2$. Choose a component $C$ as follows: if some component has at least three vertices, take such a component; otherwise take any nonsingleton component. Fix $a \in C$. For each other vertex $x$, deleting $a$ and $x$ empties at most one component. Lemma 2.1 shows that $ax$ is an edge of $\Delta_d^t(G)$. The edges from $a$ form a spanning tree of the 1-skeleton, so the fundamental group is generated by the triangular loops

$$a - x - y - a \tag{4.4}$$

for edges $xy$ outside that tree.

Suppose first that $|C| \geq 3$. Choose a neighbor $b$ of $a$ in the connected graph $C$. If $axy$ is a face, (4.4) is filled. Otherwise $b \notin \{x, y\}$. The triples $abx$ and $aby$ are faces because deleting either triple empties at most one component. Since $xy$ is a face, choose an independent $d$-set $S$ in $G - \{x, y\}$. The failure of $axy$ forces $a \in S$. Since $ab$ is an edge of $G$, $b \notin S$, so $S$ also witnesses that $bxy$ is a face. The three triangles $abx$, $aby$, and $bxy$ form a disk with boundary (4.4).

It remains to consider the case in which every nonsingleton component is $K_2$. Since $e \geq 2$, there are at least two such components. Write one as $\{a, b\}$ and choose a vertex $c$ in another. If $axy$ is not a face while $xy$ is an edge, then $\{x, y\} = \{b, s\}$ for a singleton component $s$. Indeed, without $b$ the deletion cannot both destroy the $a$-component and leave too few other components, while with $b$ the remaining vertex must exhaust a singleton. In this remaining case, $abc$, $acs$, and $bcs$ are faces and form a disk with boundary $a - b - s - a$.

Every generator (4.4) is null-homotopic. Thus the complex is simply connected. Corollary 3.4 and the wedge criterion complete the proof. □

**Proposition 4.2** (The layer $k = d+2$). *If $k = d+2$, then*

$$\Delta_d^t(G) \simeq \bigvee_{\binom{d+1}{2}} S^{n-d-1}. \tag{4.5}$$

*Proof.* Deleting two vertices empties at most two components, so Lemma 2.1 gives a complete 1-skeleton. If every component is a singleton, the complex is the complete graph on $d+2$ vertices and has cycle rank $\binom{d+1}{2}$.

Otherwise fix a vertex $v$ in a component having at least two vertices. For any two other vertices $x, y$, the deletion of $v, x, y$ cannot empty three components: emptying the component of $v$ consumes one of $x, y$, leaving only one vertex to empty another component. Therefore every triangle $vxy$ is a face. Each loop in the complete 1-skeleton is filled by a triangular fan with apex $v$. The complex is simply connected, and (4.5) follows from Corollary 3.4 and the wedge criterion. □

**Proposition 4.3** (The range $k \geq d+3$)**.** *If $k \geq d+3$, then*

$$\Delta_d^t(G) \simeq \bigvee_{\binom{k-1}{d-1}} S^{n-d-1}. \tag{4.6}$$

*Proof.* Deleting any three vertices empties at most three components, leaving at least $d$ nonempty components. Hence every three vertices form a face by Lemma 2.1. The complex contains the full 2-skeleton of the simplex on $V(G)$ and is therefore simply connected. Since $q \geq 2$, Corollary 3.4 and the wedge criterion give (4.6). □

The point of Propositions 4.1–4.3 is that no component topology enters the connectivity argument. All component hypotheses were already consumed by the homology calculation in Section 3.

## 5 The boundary layer $k = d$

When the number of components equals $d$, the target multiplicity is one, but the 1- and 2-skeleta need not be complete. We use all distributions of an independent $d$-set among the components.

Let

$$W_d = \{\mathbf{r} = (r_1, \ldots, r_d) \in \mathbb{Z}_{\geq 0}^d : r_1 + \cdots + r_d = d\} \tag{5.1}$$

and define

$$X_{\mathbf{r}} = \Delta_{r_1}^t(G_1) * \cdots * \Delta_{r_d}^t(G_d). \tag{5.2}$$

**Lemma 5.1.** *For $k = d$,*

$$\Delta_d^t(G) = \bigcup_{\mathbf{r} \in W_d} X_{\mathbf{r}}. \tag{5.3}$$

*Proof.* If $\sigma \in \Delta_d^t(G)$, choose an independent $d$-set $S \subseteq V(G) - \sigma$ and put $r_i = |S \cap V(G_i)|$. Then $\mathbf{r} \in W_d$ and the part of $\sigma$ in $V(G_i)$ lies in $\Delta_{r_i}^t(G_i)$. Hence $\sigma \in X_{\mathbf{r}}$. Conversely, if $\sigma \in X_{\mathbf{r}}$, choose an independent $r_i$-set in each $G_i - (\sigma \cap V(G_i))$; their union witnesses $\sigma \in \Delta_d^t(G)$. □

The all-ones block is

$$B = X_{(1,\ldots,1)} = \partial\Delta^{V(G_1)} * \cdots * \partial\Delta^{V(G_d)} \cong S^q. \tag{5.4}$$

Every other weak composition has both a zero coordinate and a coordinate at least 2. After void blocks are omitted, the zero coordinate supplies a full simplex factor, so every nonbaseline block is a cone.

**Lemma 5.2.** *Let $T$ be a nonempty finite set of nonbaseline weak compositions, and put*

$$m_i = \max_{\mathbf{r}\in T} r_i.$$

*If*

$$Z_T = \bigcap_{\mathbf{r}\in T} X_{\mathbf{r}}$$

*is nonvoid, then both $Z_T$ and $B \cap Z_T$ have nonempty path-connected geometric realizations. Moreover, $Z_T$ is nonvoid if and only if $B \cap Z_T$ is nonvoid.*

*Proof.* The total-cut filtration decreases with its index, so

$$Z_T = \Delta^t_{m_1}(G_1) * \cdots * \Delta^t_{m_d}(G_d), \tag{5.5}$$

whereas

$$B \cap Z_T = \Delta^t_{\max(1,m_1)}(G_1) * \cdots * \Delta^t_{\max(1,m_d)}(G_d). \tag{5.6}$$

Passing from (5.5) to (5.6) only replaces a full simplex at a zero coordinate by the boundary of that simplex. Such a boundary is nonvoid; for a one-vertex component it is $\{\varnothing\}$, the join identity. Thus the replacement does not change whether the join is void.

Some $m_j$ is at least 2. In the nonvoid case, $\Delta^t_{m_j}(G_j)$ is integer-acyclic and therefore has a nonempty path-connected realization. Its join with the remaining nonvoid factors is path-connected. This proves the assertion for $B \cap Z_T$. If all $m_i \geq 1$, then $Z_T = B \cap Z_T$. If some $m_i = 0$, then $Z_T$ has a full simplex factor and is a nonempty cone. □

**Proposition 5.3.** *If $k = d$, then $\Delta^t_d(G) \simeq S^{n-d-1}$.*

*Proof.* Enumerate the nonvoid nonbaseline blocks as $X_1, \ldots, X_s$, and let

$$U_0 = B, \qquad U_j = B \cup X_1 \cup \cdots \cup X_j. \tag{5.7}$$

For each $j$, the intersection $X_j \cap U_{j-1}$ is path-connected. Indeed, $A_j = X_j \cap B$ is nonvoid and path-connected by Lemma 5.2. Each nonvoid pairwise intersection $X_j \cap X_h$ is also path-connected, and

$$(X_j \cap X_h) \cap A_j = B \cap X_j \cap X_h$$

is nonvoid and path-connected. Consequently all pieces of $X_j \cap U_{j-1}$ are joined through $A_j$.

Every $X_j$ is contractible. Finite subcomplex inclusions are cofibrations, so the CW-subcomplex form of van Kampen applies to each attachment $U_j = U_{j-1} \cup X_j$. Since the attaching intersection is path-connected, $\pi_1(U_j)$ is a quotient of $\pi_1(U_{j-1})$. Therefore, if $q \geq 2$, the initial sphere $U_0 = B = S^q$ is simply connected and so is every $U_j$.

Corollary 3.4 gives exactly one copy of $\mathbb{Z}$ in degree $q$ and no other reduced homology. For $q \geq 2$, the wedge criterion gives $\Delta^t_d(G) \simeq S^q$. If $q = 1$, the complex is a connected graph with first Betti number one and hence is homotopy equivalent to $S^1$. If $q = 0$, it is zero-dimensional with reduced $H_0 \cong \mathbb{Z}$, hence consists of two points. The case $q = -1$ is $G = dK_1$ and gives $\{\varnothing\} = S^{-1}$ directly. □

# 6 Proof of Theorem A

Proposition 5.3 proves the case $k = d$. Proposition 4.1 proves $k = d + 1$, Proposition 4.2 proves $k = d + 2$, and Proposition 4.3 proves $k \geq d + 3$. These ranges exhaust all $k \geq d$.

The proof also explains why no single connectivity argument covers all component counts. The full 2-skeleton appears only at $k \geq d+3$; at $k = d+2$ a selected triangular fan suffices; at $k = d + 1$ one fills generators relative to a spanning star; and at $k = d$ one must use the complete weak-composition cover.

# 7 Acyclicity cannot be omitted

The component condition in Theorem A is not merely an artifact of the proof.

**Proposition 7.1.** *Let $G = C_4 \sqcup K_1$ and $d = 2$. Then*

$$\widetilde{H}_1(\Delta_2^t(G);\mathbb{Z}) \cong \mathbb{Z}, \qquad \widetilde{H}_2(\Delta_2^t(G);\mathbb{Z}) \cong \mathbb{Z}, \tag{7.1}$$

*and all other reduced homology groups vanish. In particular, the conclusion of Theorem A would predict only $S^2$, so it fails without the component acyclicity hypothesis.*

*Proof.* Write the cycle vertices in order as $a, b, c, d$ and let $z$ be the isolated vertex. The faces not containing $z$ form the boundary of the tetrahedron on $\{a, b, c, d\}$, hence an $S^2$. A face containing $z$ must leave one of the opposite independent pairs $\{a, c\}$ or $\{b, d\}$ in its complement. Therefore the remaining part is

$$z * (\Delta^{\{a,c\}} \sqcup \Delta^{\{b,d\}}),$$

attached to the tetrahedral boundary along the two disjoint edges $L = \Delta^{\{a,c\}} \sqcup \Delta^{\{b,d\}}$. Mayer–Vietoris gives one new first homology generator while preserving the fundamental 2-class, which is exactly (7.1). More explicitly, the union is the homotopy cofiber of the inclusion $L \hookrightarrow S^2$. This inclusion is null-homotopic, so the union is homotopy equivalent to $S^2 \vee \Sigma L \simeq S^2 \vee S^1$. Here $\Delta_2^t(C_4)$ is the disjoint union of two edges and is not integer-acyclic. □

This example does not prove that every individual acyclicity assumption in Theorem A is necessary. It only shows that one cannot delete all component topology and retain the same universal formula.

# 8 Relation to previous work and scope

The composition poset, its order-complex calculation, and the homotopy-colimit model (3.3) are due to Carnero Bravo [2, Proposition 12, Lemma 25, and Corollary 27]. Homotopy colimits and their spectral sequences are standard tools; see [5, 6]. The new use made here is that objectwise integral acyclicity, rather than objectwise contractibility, is enough to calculate the integral homology of the global bounded-independence complex.

The low-component proofs in Section 4 and the contractible-factor version of the weak-composition cover first appeared in the author's preprint [8], which answered [2, Question

29]. The present Section 5 replaces the former deformation-retraction argument by connected intersections and van Kampen, so that the same layer also works under integer acyclicity. The weak-composition cover is related in spirit to the two-factor decomposition in [2, Theorem 34] and to polyhedral-join methods [7], but Proposition 5.3 is not obtained by iterating the two-factor theorem: intermediate unions need not retain its null-homotopy hypotheses.

Theorem A subsumes the affirmative answer to [2, Question 29] in [8] and answers [2, Question 30] under the component total-cut assumptions appearing there. More precisely:

- it covers every $d \geq 2$ and every $k \geq d$;
- it removes the condition that each $\mathrm{BI}_2(G_i)$ be simply connected;
- it replaces "void or contractible" by "void or integer-acyclic" for each $\Delta_r^t(G_i)$, $2 \leq r \leq d$.

It does not classify arbitrary disconnected graphs. Proposition 7.1 shows that arbitrary components can add homology outside the predicted degree. The paper also does not determine simple-homotopy type, shellability, torsion when the hypotheses fail, or the maps between consecutive total-cut filtration levels. No finite computation is used to prove Theorem A.

## Acknowledgment of AI assistance

AI tools were used for literature discovery, proof stress-testing, computational falsification during exploration, and drafting. Carptopus is the responsible author and is solely responsible for the mathematical claims and the decision to disseminate this manuscript.

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